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Multiscale recoverable information in endogenous A-to-I RNA editing

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Endogenous adenosine-to-inosine RNA editing depends on duplex formation, cellular context, and nucleotide choice, yet how much each scale contributes remains unresolved. We used controlled held-out prediction to assay what could be recovered from measured variables across 884 human inverted-Alu substrates. Whole-duplex sequence and predicted structure predicted editing across human endpoints, with less consistent transfer to heterogeneous nonhuman datasets. Genomic inter-arm geometry accounted for additional between-substrate variation in editing burden, although it did not reliably improve site-level predictions. The same genomic sites often varied across tissues. Available RBP, transcript, repeat, and bulk chromatin annotations nevertheless supplied no reproducible residual signal at the site level. Pair-first weighting improved learning from correlated sites. A local representation centered on the target and organized by predicted pairing added information beyond whole-duplex pooling, whereas shifting the representation away from the target erased the gain despite matched capacity. Richer supervision and model expansion produced limited or adverse returns under the tested regime. ADARPro, a frozen predictor combining the supported principles, improved AUROC, AUPRC, F1, and Brier score in all five held-out endpoints. Substrate- and nucleotide-scale signals are recoverable, whereas current bulk measurements leave site-level variation unresolved.

Significance

A-to-I RNA editing helps cells tolerate endogenous double-stranded RNA, yet why one RNA duplex or adenosine is edited while another is not remains unclear. We used controlled prediction to test measured information at three biological scales. Whole-duplex sequence and predicted structure recovered broad editing patterns. Genomic organization added information about substrate-level editing propensity, and local context centered on the target and its predicted pairing added complementary site-level information. Bulk tissue and cell-line annotations supplied no reproducible additional signal at the site level. The resulting map identifies what current data can predict and motivates measurements resolved by donor, cell, transcript, and RNA structure.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA). This reaction diversifies transcripts and prevents endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Potential substrates are not uniformly edited. Nearby adenosines in one predicted helix can differ sharply, and the same locus can differ among tissues (11, 12). Local sequence preference, mismatches, bulges, and helix context contribute to this selectivity (13–17). They do not specify whether a transcript forms and retains the relevant duplex, occupies the appropriate compartment, or encounters active ADAR.

Three biological scales may contribute to this selectivity. Whole-duplex sequence and folding constrain intrinsic substrate suitability. Transcript- and cell-specific state may govern whether that substrate forms and remains accessible, whereas local context distinguishes adenosines within an available duplex. How much each scale contributes has not been compared under one predictive design. It remains unclear which editing patterns can be recovered from existing measurements, which are obscured by measurement scale, and when additional model complexity stops helping.

Controlled held-out prediction cannot establish mechanism. Under a specified representation, sample size, and evaluation regime, it can test whether a measured variable improves prediction beyond

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a fixed sequence–structure reference. Such a gain identifies reproducible signal absent from that reference. A null result applies only to the tested measurement and formulation and may reflect redundancy, noise, finite data, or model misspecification. Comparisons, ablations, transfer tests, and saturation experiments can delimit what present data encode without treating association as causality.

AdarEdit showed that a compact graph representation of sequence and predicted RNA structure can predict site-level editing in a pair-partitioned benchmark (18). We used the benchmark and model as a common reference for a different question: which biological scales account for that predictive signal, what adds to it reproducibly, and when more measurements or model complexity stop helping. Whole-duplex inputs recovered broad intrinsic signal. Genomic organization added pair-level burden information, and a target-centered local representation recovered signal lost through global pooling. Bulk contextual measurements did not close the remaining gap. A final held-out evaluation tested a predictor that combined the supported principles.

Results

Editing can be interrogated at substrate, cellular, and target scales. We distinguished the intrinsic suitability of an RNA duplex, its transcript and cellular context, and selection of an adenosine within it (Fig. 1A). Separate comparisons tested whole-duplex sequence and predicted structure, substrate organization, available contextual measurements, target-centered representations, and the returns from additional supervision or model capacity.

We analyzed 884 graph-buildable inverted-Alu substrates across four individual GTEx tissues and Combined, a pooled endpoint. Complete Alu pairs were assigned without overlap to 566 training, 139 validation, and 179 test pairs. Discovery and model selection used pair-disjoint out-of-fold comparisons with pair-clustered uncertainty; the complete predictor and analysis manifest were frozen before the canonical held-out evaluation (Fig. 1B and C). Label definitions, fold isolation, evidence-status criteria, and evaluation safeguards are detailed in Methods and SI Appendix.

Sequence and predicted structure recover broad but non-universal signal. The compact sequence–structure baseline achieved within-context AUROC of 0.869–0.933 across the five human endpoints and 0.823–0.865 across three nonhuman datasets (Fig. 2A). The datasets differ and these ranges do not estimate a common effect. They nonetheless show that whole-duplex sequence and predicted structure predict editing beyond one species or tissue.

Transfer was strongest among human endpoints. It varied more among nonhuman datasets, and human–nonhuman transfer was lower and asymmetric (Fig. 2B and C). Sequence–structure regularities extend across contexts without being universal. These datasets cannot separate the effects of lineage, substrate composition, assay, labeling, and cellular environment.

High discrimination does not imply biological completeness. A sequence–structure model can rank sites while leaving substrate availability, tissue state, and unmeasured conformations unresolved. The remaining comparisons tested what could be added beyond this baseline; predictive accuracy was not treated as explained biological variance.

Substrate organization predicts variation in editing burden. Genomic separation between Alu arms correlated negatively with edited-site fraction in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 3B). This distance is absent from the site-level graph input, which joins the arms with a fixed separator. Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 3C; SI Appendix). Substrate organization provides a reproducible pair-level signal beyond the tested graph summaries. The modeled outcome is editing propensity; duplex realization and ADAR engagement were not measured.

Tissue variation is evident, but current bulk measurements do not recover it. Editing outcomes varied among tissues even when genomic site identity was fixed. In a post-hoc analysis of development pairs, 1,221 of 74,758 sites observed in at least two individual tissues switched strictly between edited ($\geq 15\%$) and nonedited ($< 1\%$) states. Among 50,162 sites observed in all four tissues, 17.4% spanned at least 15 EditingIndex points. Alu-pair rankings on identical site sets still correlated across every tissue pair ($\rho = 0.727$–0.921 for mean EditingIndex; SI Appendix, Fig. S7). Stable substrate differences coexist with tissue-dependent site outcomes.

Predicted substrate burden did not translate reliably into better site-level classification. A cross-fitted gate added 0.0084 AUPRC but only 0.0041 F1, whose interval included zero, and increased Brier score by 0.0051. A neural burden auxiliary also failed its prespecified criterion. Static geometry captures pair-level propensity without identifying why individual sites depart from it.

The tested contextual measurements added no reproducible residual signal at the site level. They comprised replicated ENCODE eCLIP occupancy, sequence-based RBP predictions, tissue-average abundance and their interactions, with randomized or shuffled controls (19–22). Sequence-based predictions complemented eCLIP, for which Alu multimapping makes peak absence ambiguous. Neither cell-line occupancy nor tissue-average abundance reproducibly improved the reference. Tissue identity alone nearly separated cross-tissue label switches, leaving four tissue means unable to identify a transferable RBP-specific effect. This negative result applies to the tested proxies and does not conflict with established RBP regulation (23–26).

Transcript and repeat annotations also failed to improve prediction beyond the graph. Exact-tissue bulk H3K36me3 overlap produced a small liver AUPRC increment without stable AUROC, F1, or pair-level benefit (27, 28). Across these tests, static, cell-line, and tissue-average measurements did not recover the site-level variation left after substrate propensity was modeled. The source of that variation, whether cellular state, measurement noise, finite data, or representation error, remains unresolved.

Pair-first weighting prevents site-rich substrates from dominating learning. Sampling substrates before sites improves pre-

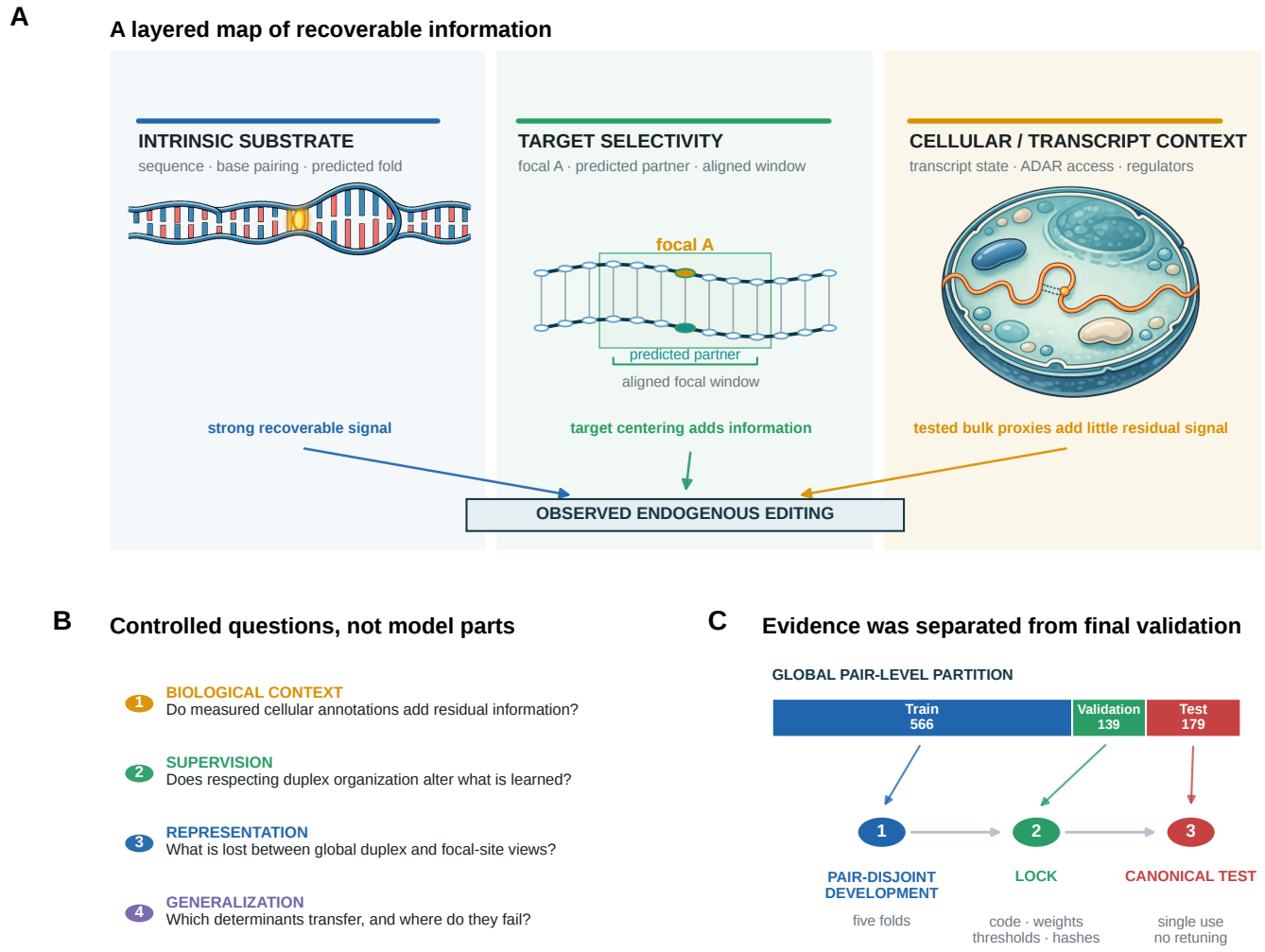


Fig. 1. Study design separates biological scales from held-out evaluation. (A) Intrinsic substrate properties, target-centered local context organized by predicted pairing, and transcript/cellular context may contribute to observed editing; pair burden is an indirect measure of substrate propensity. Illustrations are conceptual. (B) Controlled comparisons test context, supervision, representation, and generalization. (C) Complete pairs were partitioned globally, development used pair-disjoint folds, and the final specification was frozen before held-out evaluation. Combined is pooled and received no tissue-specific state vector.

diction independently of adding labels. The canonical row set contains only part of the available high- and low-editing observations. Under uniform site sampling, a substrate's influence grows with its number of observations even though sites in one duplex share sequence, structure, genomic context, and often editing behavior.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 4A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that expanded pool fixed and drawing a pair before class and site contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). Label recovery and substrate weighting made separable contributions.

The pair-first formulation improved the same-fold reference in all five endpoints (Fig. 4B); all 15 AUROC, AUPRC, and F1 intervals excluded zero. Within this benchmark, sampling pairs uniformly prevents site-rich substrates from receiving disproportionate influence during learning.

Pair-first learning improved log loss most for pairs with the fewest eligible sites, whereas the two largest site-count quintiles favored row-wise learning (SI Appendix, Fig. S8B).

Target-centered local information improves site-level prediction. Whole-duplex pooling loses some target-specific signal. Simple target-contact geometry improved ranking and calibration but not F1. Explicitly retaining the focal graph node improved AUROC and AUPRC more consistently than an otherwise matched duplicate-global control. Both low-capacity comparisons placed the additional signal near the target before a specialized local encoder was built.

The selected local-augmented graph predictor uses a 65-position representation centered jointly on the focal adenosine and its dot-bracket-aligned helix partner. A low-capacity dilated convolution encodes nucleotide identity, pairing state, focal and partner indicators, and signed relative position. Its output is fused with the whole-duplex graph embedding before classification. In muscle OOF data, the model reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared

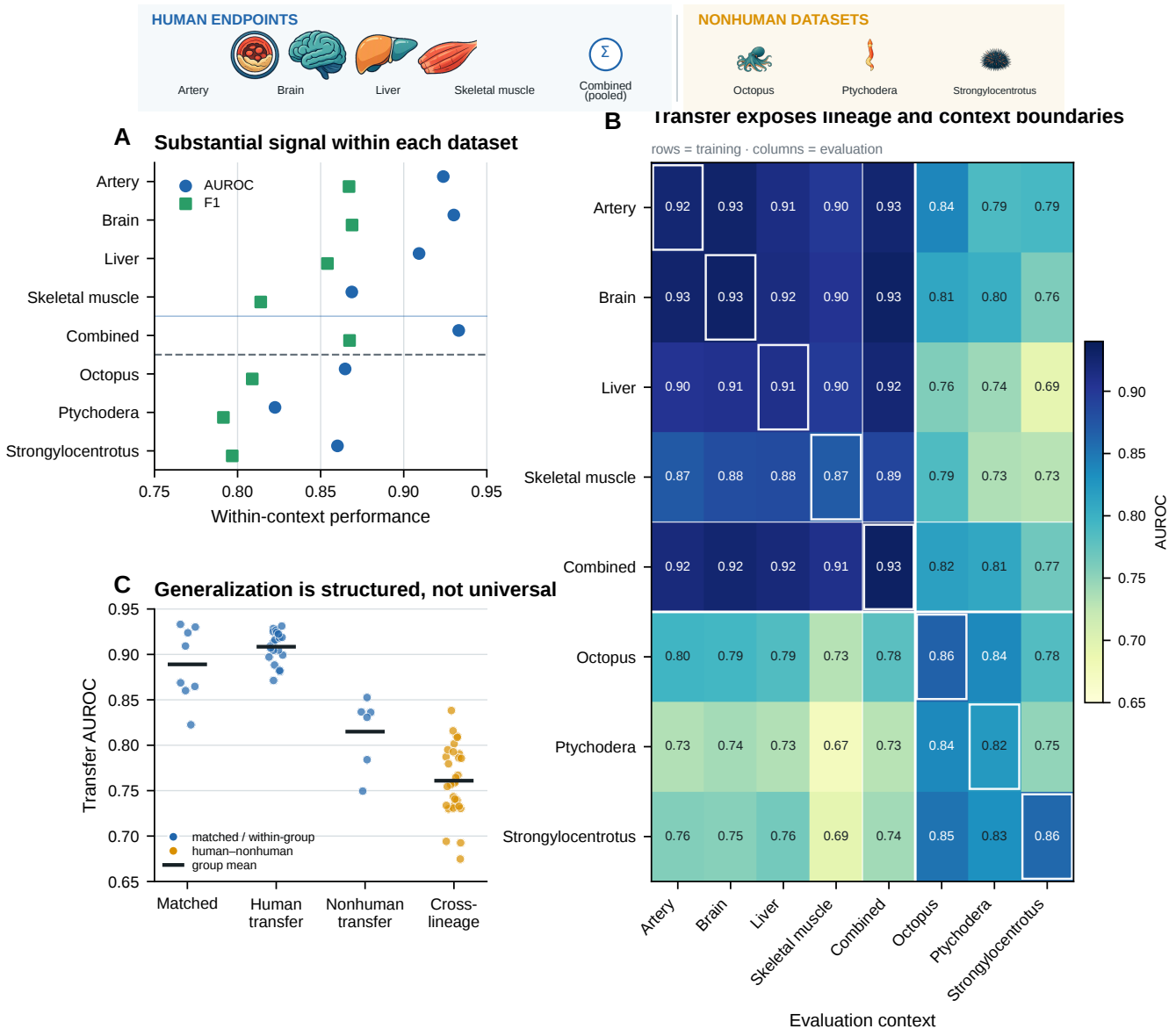


Fig. 2. Sequence and predicted structure carry substantial but non-universal signal. The tissue and lineage illustrations identify dataset classes and encode no measured values. (A) Within-context AUROC and F1 for the retained compact sequence–structure baseline across five human endpoints and three nonhuman datasets. Combined is pooled. (B) AUROC when each training context is evaluated in every context; white lines separate human and nonhuman blocks, and outlined diagonal cells mark matched-context evaluations. (C) Transfer grouped by the relation between training and evaluation contexts; horizontal bars are group means. Dataset heterogeneity precludes causal attribution or meta-analysis.

with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 4C).

A matched-capacity control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to 0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. The gain depends on the correctly centered, pairing-organized neighborhood, not simply on added convolutional capacity. Because the full window was shifted, the control does not isolate the predicted partner from the rest of that neighborhood. Liver replication and the secondary natural-prevalence analysis showed the same pattern.

In a descriptive structural stratification, Brier score improved for paired and unpaired targets and across every target-stem extent (SI Appendix, Fig. S8C and D). The gain cannot be reduced to a binary paired-status annotation or one stem-length threshold.

Whole-duplex and target-local information remain complementary. The whole-duplex and local-augmented predictors made sufficiently different errors to benefit from combination. A fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver. All six pair-clustered intervals excluded zero (Fig. 4D).

The final predictor uses two independently trained pair-first components: a whole-duplex-only graph predictor and a graph predictor augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed 0.5/0.5 mean, with no weight search

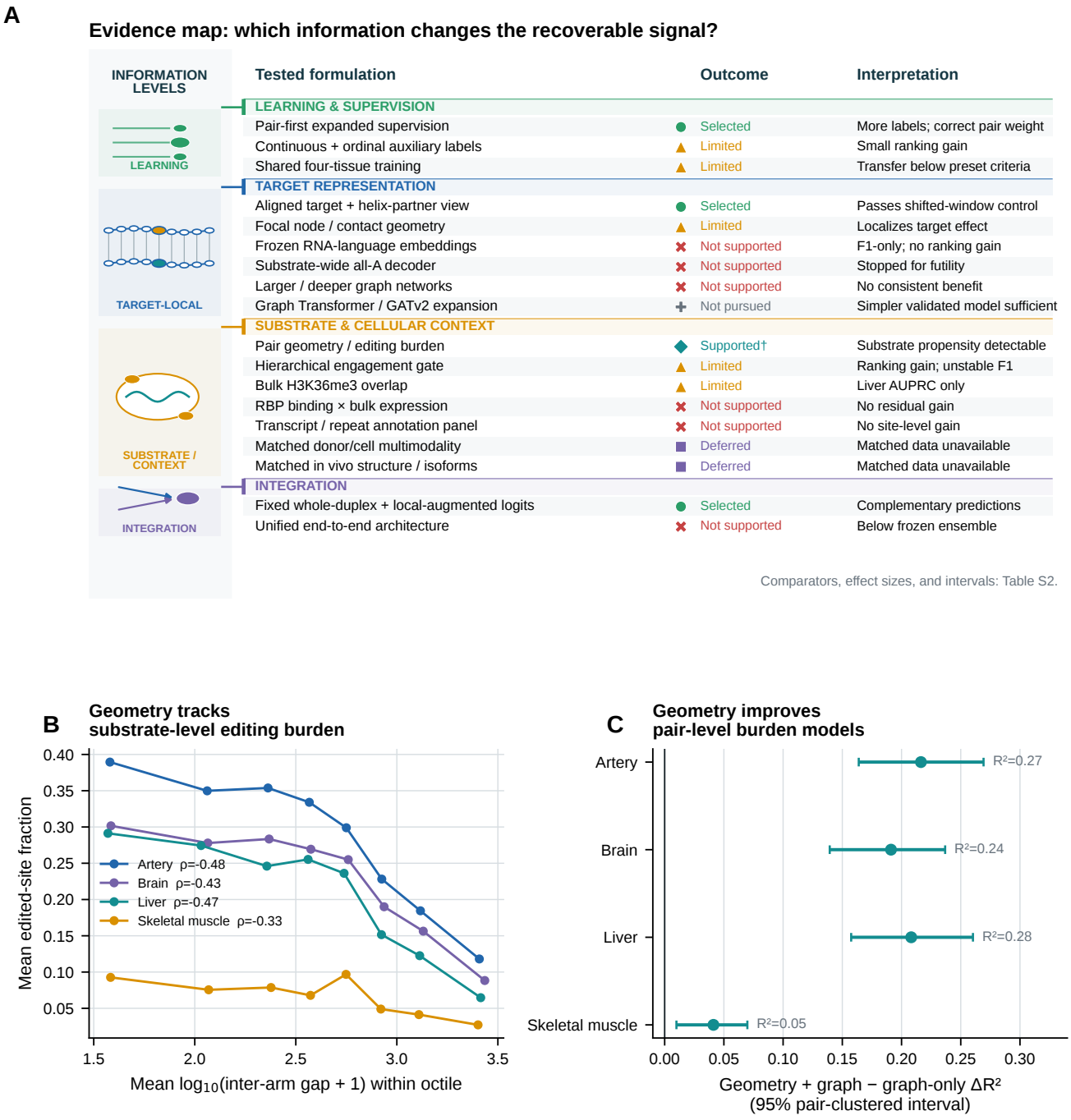


Fig. 3. Genomic organization predicts substrate-level editing burden. (A) Tested measurements grouped by scale and outcome; the left hierarchy corresponds to the Results sections. SI Appendix, Table S2 gives each comparator, effect size, and interval. "Supported†" denotes a reproducible substrate-level observation not yet useful for site prediction; "not supported" applies only to the tested formulation, and deferred measurements were unavailable. (B) Mean edited-site fraction across octiles of inter-arm gap in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 after adding genomic geometry to graph summaries. Bars are 95% pair-clustered bootstrap intervals; labels give absolute R^2 for the combined model.

(Fig. 4E). This fixed combination retains both predictive views without fitting a flexible stacker to only hundreds of independent substrates.

In a post-hoc equal-pair analysis restricted to duplexes containing both labels, the fixed ensemble improved mean within-duplex AUROC over the pair-first component in muscle (+1.78 points; 95% pair-bootstrap interval +1.21 to +2.40) and liver (+1.38 points; +1.06 to +1.70). The aligned-local component alone showed a clear within-duplex increment only in muscle, so this analysis supports complementary within-substrate ranking rather than a replicated local-component effect (SI Appendix, Fig. S8A).

Richer supervision and model complexity yield diminishing returns. With substrate-aware sampling and target-centered local information already represented, graded supervision and cross-tissue sharing produced only small ranking gains and

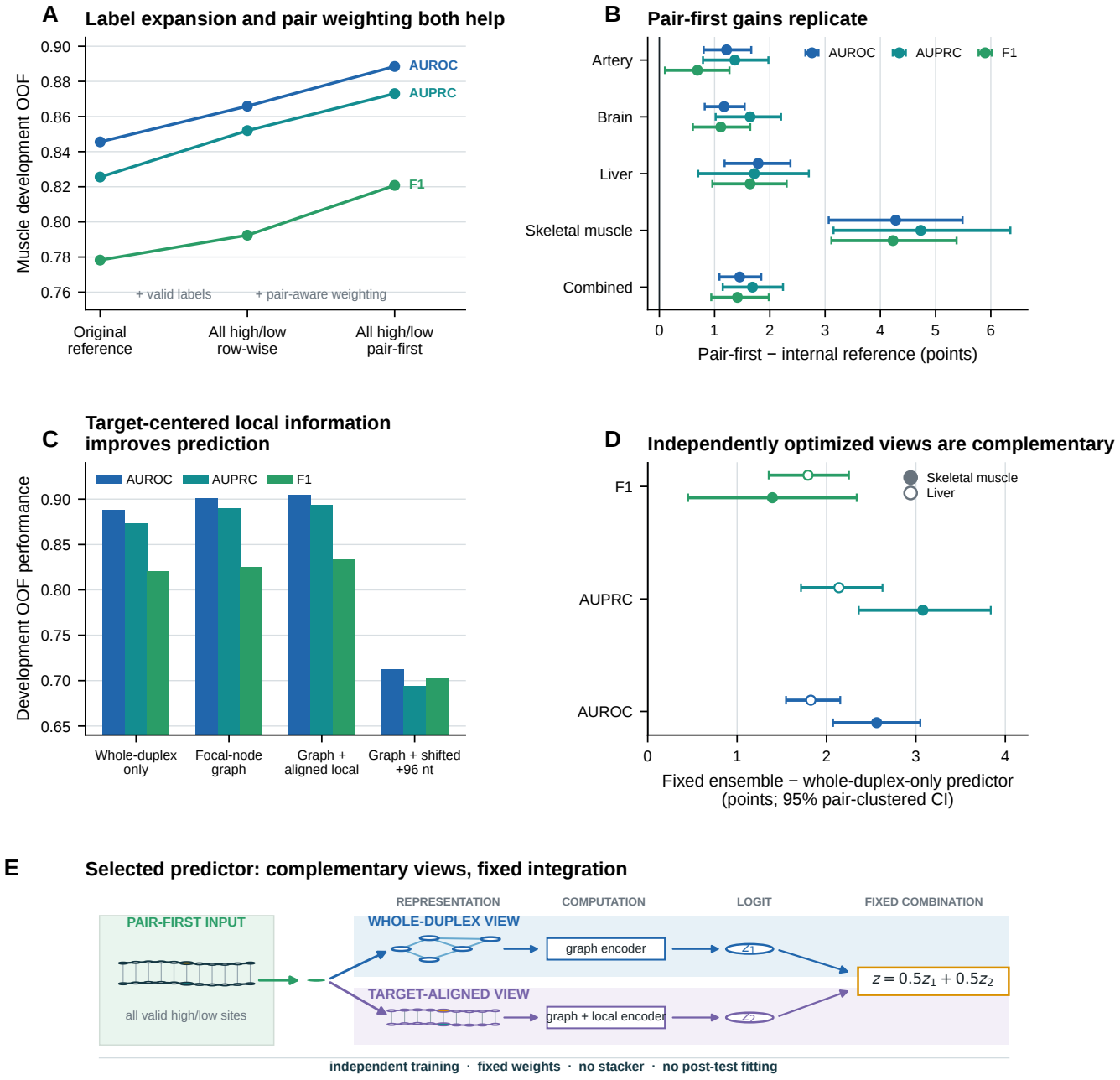


Fig. 4. Pair-first learning and target-centered representation provide complementary gains. (A) Same-fold ablations separate recovery of additional valid high/low labels from pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, focal-state retention, and aligned-local fusion form an evidence ladder; an otherwise identical local model is shifted by 96 nt. (D) A fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) The predictor combines independently trained whole-duplex-only and graph-plus-local models without searched weights or post-test fitting. The RNA illustration is conceptual. All panels show development evidence.

adverse threshold or calibration tradeoffs. Both fell below their predeclared selection criteria (Fig. 3A; SI Appendix).

Substrate-wide decoding, frozen RNA-language embeddings, larger graph networks, and a preregistered connected integration were also neutral or adverse in their prespecified comparisons (29) (SI Appendix). Under the tested measurements, representations, and effective sample size, performance had saturated. This conditional result is not a universal prediction ceiling.

A frozen predictor generalizes to unseen benchmark pairs. After development selection, all predictor components, checkpoints, thresholds, weights, code, and hashes were frozen before canonical held-out evaluation. ADARPro improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 5A), with mean gains of 0.0277, 0.0261, and 0.0253 over reproduced supplied-checkpoint AdarEdit. Every endpoint had a positive 95% pair-clustered interval for all three metrics, and Brier score decreased throughout (Fig. 5B and C; SI Appendix, Table S1).

Combined overlaps the individual tissue endpoints, so the mean is descriptive rather than a five-tissue meta-analysis. The frozen combination generalizes to unseen pairs within this benchmark. Its component-

level attribution rests on the controlled development comparisons above.

At frozen validation-selected thresholds, ADARPro corrected 32.0–50.6% of baseline false positives while converting 1.7–5.9% of baseline true positives into false negatives (Fig. 5D). A secondary matched-recall analysis is reported in SI Appendix, Table S9.

Discussion

Predictive signal in endogenous A-to-I editing separates across substrate and nucleotide scales. Whole-duplex sequence and predicted structure predict broad patterns of intrinsic suitability. Genomic organization marks a reproducible substrate-level editing propensity, whereas target-centered local context organized by predicted pairing adds signal obscured by whole-duplex pooling. Pair-first learning addresses a separate statistical problem: correlated sites should not gain influence simply because one substrate contributes more rows.

Prior work identified sequence preferences and helix context as determinants of local editability, while the distance between oppositely oriented Alus predicts aggregate Alu editing (13, 16, 17). Perturbation experiments have also established regulation by RBPs and chromatin-associated processes (23–27). The present analysis places these observations at different predictive scales. Intrinsic RNA and target context are recoverable from the benchmark, but cell-line occupancy and tissue-average abundance or chromatin do not identify regulator exposure or substrate state on an individual transcript. The null results apply to the tested measurements, not to the underlying RBP or chromatin biology.

Performance gains diminished after the models represented whole-duplex and target-centered information and accounted for substrate dependence. Added supervision, generic RNA representations, greater capacity, and joint integration were neutral or adverse under the tested regime. Residual error may arise from donor- or cell-specific ADAR state, transcript processing, regulator occupancy, in vivo conformation, measurement error, finite data, model misspecification, or biological stochasticity. These experiments cannot apportion those sources. They locate a gap between current bulk measurements and site-level outcomes without assigning it to a single latent cellular state.

The scope is limited in several ways. The primary benchmark contains graph-buildable human inverted-Alu substrates and tissue summaries, not all editing substrates or donor-resolved measurements. Heterogeneous nonhuman datasets show transfer loss but confound lineage, substrate class, assay, and labeling effects. Predicted secondary structure is a surrogate for the conformations available to ADAR in vivo. The canonical task is balanced and excludes intermediate observations; natural-prevalence analyses are secondary, and eCLIP coverage is cell-line-specific and repeat-censored. The frozen test assesses unseen pairs from the same benchmark construction, whereas component attribution comes from adaptively selected, nonmultiplicity-adjusted development comparisons. The upstream study reports cross-cohort label concordance, but the present predictor was not evaluated on an independently generated human cohort. Its transportability remains untested (18).

The frozen predictor combined the supported information scales and improved held-out discrimination, classification, and calibration. Resolving the remaining variation will require editing measurements paired with transcript abundance and processing, splicing and intron retention, ADAR isoform and interferon state, enzyme activity and localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation from the same donors or cells and substrates. Such data could test whether cellular state governs realization of a suitable duplex and local RNA context governs selection of particular adenosines.

Materials and Methods

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human sites were labeled edited at EditingIndex $\geq 15\%$, nonedited at EditingIndex $< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. We verified pair identities and the fixed 566/139/179 train/validation/test assignment before analysis. Combined was used only as its published pooled endpoint.

Development protocol. All discovery used the original train and validation pairs, divided into 5 outer folds by pair ID. Learned preprocessing, feature selection, checkpoint choice, and threshold choice excluded the outer fold. We concatenated predictions only after every row had been scored by a model that was not trained on its pair. The canonical test remained inaccessible during development. Development OOF metrics estimate same-fold incremental effects; supplied-checkpoint test metrics remain the authoritative reference.

Models. The first component retained the compact AdarEdit graph attention encoder and was trained from the expanded high/low site pool using uniform pair-first and equal-class sampling. The second component was trained independently with the same graph encoder. It added a 65-position target/helix-partner-aligned representation encoded by two dilated convolutional layers, then fused the local and graph embeddings before prediction. The selected probability was $\sigma[(\operatorname{logit} p_{\text{whole}} + \operatorname{logit} p_{\text{augmented}})/2]$, with weights fixed at 0.5/0.5 and no weight search. The shifted-window control preserved the graph-plus-convolution architecture and optimization while moving the local center 96 nt. SI Appendix gives full feature definitions, architectures, ablations, and biological-context protocols.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each frozen training run. For uncertainty, a pair-clustered bootstrap with 2,000 replicates resampled Alu pairs and retained all their sites. Development intervals quantify paired effect uncertainty and were not treated as multiplicity-adjusted confirmatory tests. Selection also required recorded decision criteria and controls. Code, inputs, weights, epochs, thresholds, and checkpoint SHA-256 values were frozen before one held-out evaluation; no subsequent model selection or tuning occurred. Natural-prevalence and matched-recall analyses are secondary and described in SI Appendix.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. GPT Image 1.5 generated only label-free, nonquantitative figure components; prompts, candidate selection, and hashes are disclosed in the repository. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

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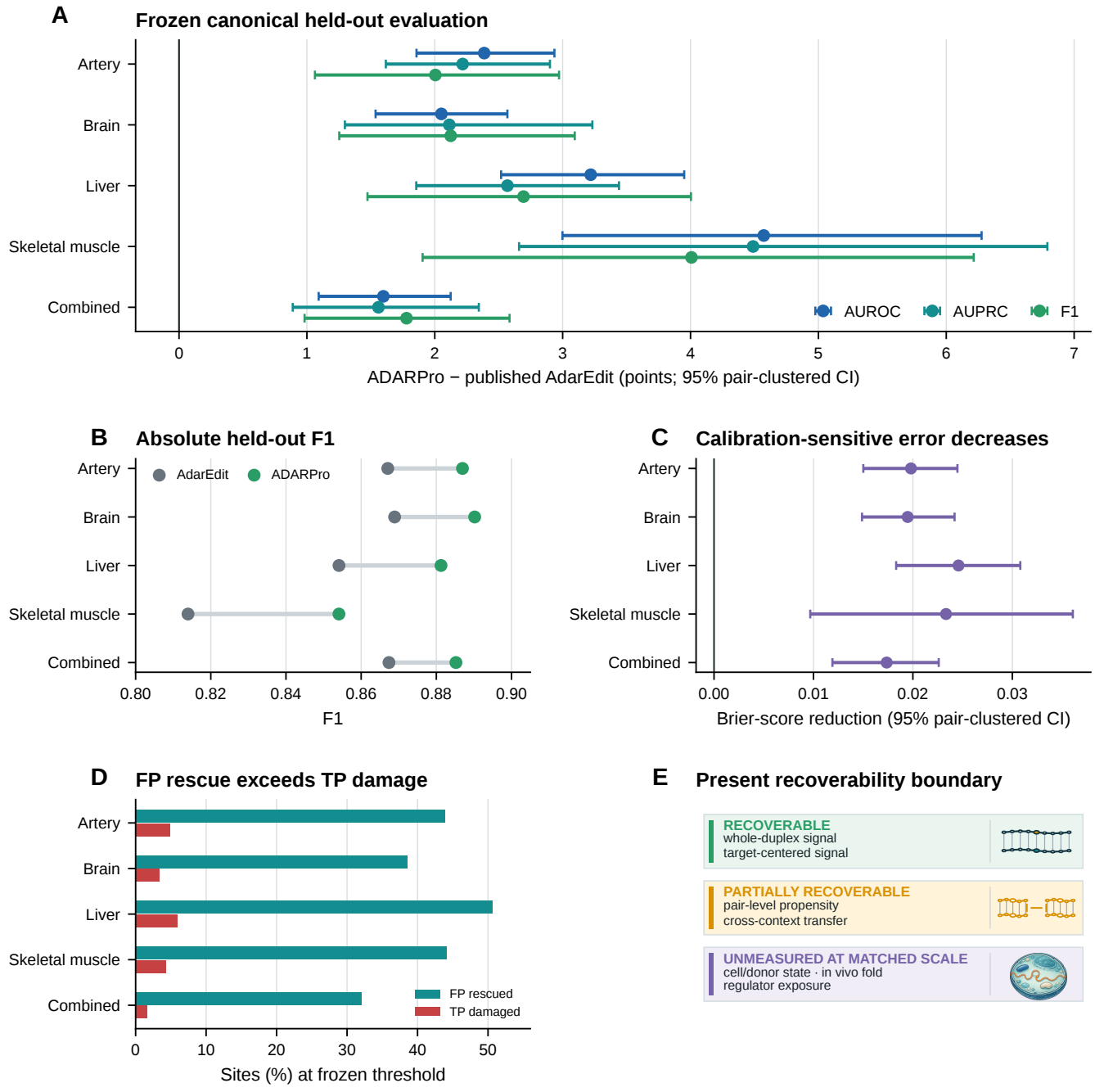


Fig. 5. Frozen evaluation confirms within-benchmark generalization and summarizes what current measurements recover. (A) ADARPro minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score. (D) Baseline false positives corrected and baseline true positives converted to false negatives at frozen validation-selected thresholds. (E) Information captured, partially resolved, and unavailable at matched biological scale; tier-specific icons are conceptual. Combined is pooled. All quantitative panels use the frozen canonical held-out evaluation.

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